1 Signal Analysis

Signal analysis is performed in this chapter. The aim is to first provide a brief introduction to the Discrete Fourier Transform (DFT) in the context of its application to multi-frequency analysis for flutter derivative identification. This is then applied to single- and multi-frequency signal extraction. Finally, potential sources of error and methods to mitigate them are discussed.

The Fourier Transform is a mathematical tool used to convert a signal from the time domain to the frequency domain. Since real-world data consists of discrete numbers rather than continuous mathematical functions, the transform cannot be applied directly. Therefore, the Discrete Fourier Transform (DFT) is used in such cases. The DFT converts time-varying data into the frequency domain, and its inputs are the time-varying data and the sampling frequency. The sampling frequency defines how many samples are taken per second; in the CFD context, it is equal to $1/\Delta t$, where Δt is the time step.

The Fast Fourier Transform (FFT) is derived from the DFT, which has significantly higher computational efficiency. Several open-source tools are available to perform FFT computations, one of which is provided by the `scipy.fft` module in Python [**?**]. Since Python is used for post-processing throughout this thesis, the SciPy FFT tool is used to convert the obtained aerodynamic forces and moments into the frequency domain.

Since the DFT operates on a discrete set of values, it is subject to certain inherent limitations that can affect the quality of the transformation and the usability of its results. It is therefore necessary to study these issues before applying the FFT to flutter derivative calculations. This is the focus of the present chapter: to examine the issues associated with the DFT and to investigate its efficient application in the multi-frequency method for flutter derivative identification.

The accuracy of extracting the amplitude of a particular signal using the FFT depends on two parameters:

1. The frequency of the signal to be extracted.
2. The frequency bin of the data.

The frequency bin refers to the discrete set of frequency values produced by the FFT during domain conversion. If the frequency of the signal to be extracted coincides with one of the values in the frequency bin, the extracted amplitude will be accurate and free from discretisation errors. If the signal frequency does not coincide with a frequency bin value, the amplitude obtained from the FFT will contain an error. This effect is known as *spectral leakage*: since no bin directly

corresponds to the signal frequency, its amplitude is distributed across the neighbouring bins. The frequency bins of a time data is calculated as

$$f_k = k \cdot \frac{f_s}{N}, \qquad k = 0, 1, 2, \ldots, \left\lfloor \frac{N}{2} \right\rfloor \tag{1.1}$$

where

- f_k = frequency of the k-th frequency bin (Hz)
- k = frequency bin index
- f_s = sampling frequency (Hz)
- N = length of the data
- $\Delta f = \frac{f_s}{N}$ = frequency resolution / bin width (Hz)

The spacing between two adjacent frequencies in the bin is known as the *bin spacing*, and the *frequency resolution* of an FFT refers to how small this spacing is. A smaller bin spacing corresponds to a higher frequency resolution. As shown by the formula, the resolution depends on the length of the data record.

Another important issue to be aware of is *aliasing*. The maximum signal frequency that can be extracted using an FFT is limited by the Nyquist frequency, given by:

$$f_{\text{Nyquist}} = \frac{f_s}{2} \tag{1.2}$$

where f_s is the sampling frequency. This can create issues in two ways. First, if the frequency of interest lies above the Nyquist frequency, it cannot be extracted. Second, even if the frequency of interest lies within the Nyquist limit, any other signal with a frequency above it will be folded back into the valid frequency range by the FFT. If such a folded component does not fall on a bin, it will introduce additional spectral leakage. While the probability of this occurrence is low, this effect should be kept in mind when using the FFT in practice.

These two effects combined impose restrictions on the frequencies that can be extracted and the accuracy with which they can be obtained, in various forms. These are studied for single- and multi-frequency methods in the following sections.

1.1 Single Frequency

This section studies the effects discussed in Section 1 as applied to a single-frequency signal. A sample sine wave is generated with discrete points, passed to the FFT tool from the SciPy module,

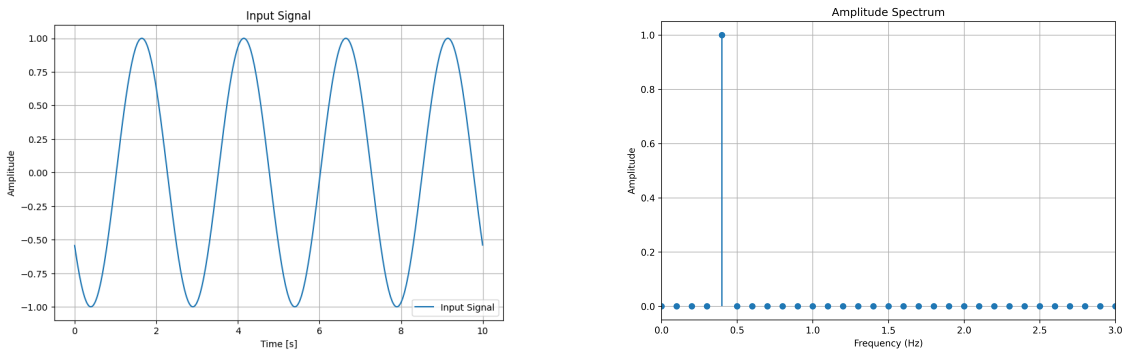
and the resulting amplitudes are analyzed. Since the amplitude of the input signal is known, it is compared against the FFT output and the error is examined for the following cases:

1. Bin Match
2. Non-Bin Match
3. Bin Match with Noise
4. Frequency Bin Resolution on Non-Matching Bin

Throughout this section, the input signal has a total duration of 10 seconds and is sampled at a frequency of 500 Hz. With these parameters, the resulting frequency bins range from 0 to 250.0 Hz, with a bin spacing of 0.1 Hz (i.e., 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, …, 250.0 Hz). The maximum frequency of 250 Hz corresponds to the Nyquist limit, which represents half the sampling frequency and is the highest frequency that can be accurately represented by the discrete sampling.

1.1.1 Bin Match

A signal with a frequency of 0.4 Hz and an amplitude of 1.0 is used as input. Since 0.4 Hz falls directly on a frequency bin, no spectral leakage is expected. As shown in Figure 1, the amplitude output from the FFT is 1.00, which matches the input signal amplitude exactly. This confirms that, when the signal frequency falls on a frequency bin, no discretisation error is introduced.



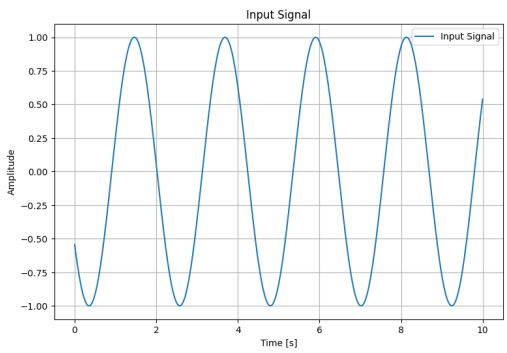
(a) Input signal at 0.4 Hz. **(b)** FFT amplitude spectrum — bin match.

Figure 1: Single-frequency bin-match case: input signal and corresponding FFT spectrum.

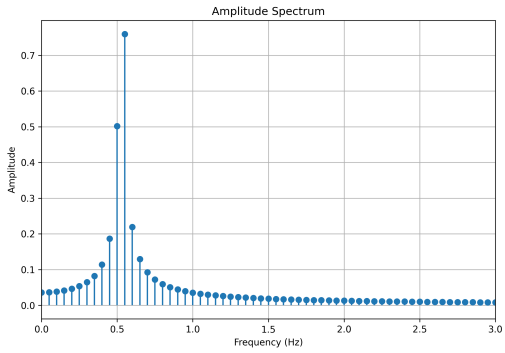
1.1.2 Non-Bin Match

In this case, the input signal frequency is changed to 0.45 Hz, which does not coincide with any frequency bin. Since 0.45 Hz lies midway between the 0.4 Hz and 0.5 Hz bins, the FFT assigns its amplitude to the nearest neighbouring bin.

As shown in Figure 2, the extracted amplitude is 0.653, corresponding to an error of 34.70%. This demonstrates that when the signal frequency does not fall on a frequency bin, spectral leakage causes a significant reduction in accuracy.



(a) Input signal at 0.45 Hz.

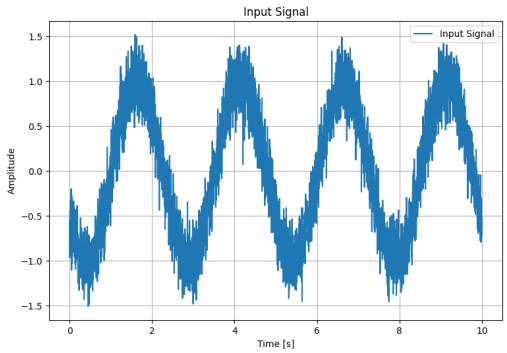


(b) FFT amplitude spectrum — non-bin match.

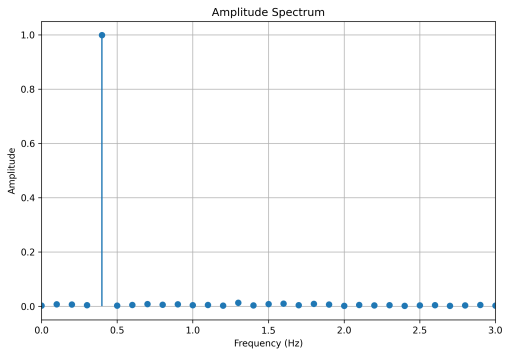
Figure 2: Single-frequency non-bin-match case: input signal and corresponding FFT spectrum showing spectral leakage.

1.1.3 Bin Match with Noise

In this case, the input signal is kept at 0.4 Hz. Additionally, white noise with an amplitude of 0.2 is superposed on the signal to assess its effect on the extracted amplitude. As shown in Figure 3, the extracted amplitude is 0.99, corresponding to an error of only 0.04%. Although noise does introduce a small error, its effect is considerably less severe than that of spectral leakage.



(a) Input signal at 0.4 Hz with white noise.



(b) FFT amplitude spectrum — bin match with noise.

Figure 3: Single-frequency bin-match case with added white noise: input signal and corresponding FFT spectrum.

1.1.4 Frequency Bin Resolution on Non-Matching Bin

The amplitude error associated with a non-bin-matched frequency is based on the relative position of that frequency between its two nearest bins. The closer the frequency lies to the midpoint between two adjacent bins, the greater the spectral leakage and, consequently, the larger the resulting error.

To illustrate this, consider a signal at 0.54 Hz with the same configuration as above, where the neighbouring frequency bins are located at 0.5 Hz and 0.6 Hz. This frequency falls near the centre of the interval between the two adjacent bins, producing an error of 21.77%. When the total time is extended to 20 s, the bin spacing is reduced, placing the neighbouring bins at 0.5 Hz and

0.55 Hz. In this finer-resolution case, the error decreases to 5.6%, since 0.54 Hz lies closer, in relative terms, to one of its neighbouring bins.

This behaviour is further illustrated by examining a signal at 0.53 Hz. In the first configuration (bin spacing of 0.1 Hz), the normalised distance of 0.53 Hz from the midpoint between its nearest bins is $0.3/0.5 = 0.6$, yielding an error of 11.68%. In the second configuration (bin spacing of 0.05 Hz), this ratio increases to $0.2/0.25 = 0.8$, and the error rises to 24.02%, despite the absolute distance to the nearest bin being smaller. This result highlights that it is the relative position within the bin interval that determines the magnitude of spectral leakage error. One approach to mitigate error in the case where the frequency falls exactly at the midpoint between two bins is to sum the amplitudes of the two neighbouring bins. However, this introduces additional uncertainty and should therefore be applied with caution.

Another approach is improving the frequency resolution through a longer data record, which reduces spectral leakage error by moving the signal frequency closer to a bin in relative terms. However, as the resolution increases, the sensitivity of the error also increases, which must be carefully considered when increasing the length.

1.2 Multi-Frequency

This section extends the analysis of the previous section to multi-frequency signals. The primary objectives are to determine whether increasing the number of frequency components affects accuracy, to examine the impact of a frequency not falling on a bin in a multi-component signal, and to assess the effect of noise in such cases.

The data length and frequency bins are kept identical to those used in the single-frequency analysis:

- Total time: 10 seconds
- Sampling rate: 500 Hz

Frequency Bin: $0 \mid 0.1 \mid 0.2 \mid 0.3 \mid 0.4 \mid 0.5 \mid 0.6 \mid \cdots \mid 250.0$ Hz

The following cases are examined:

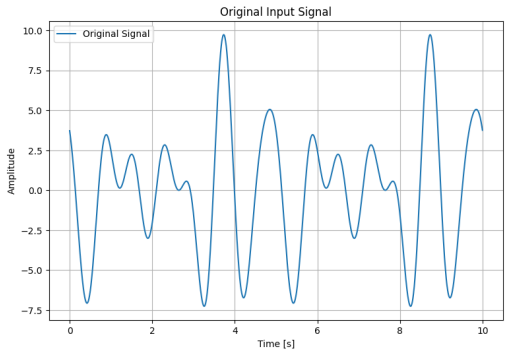
1. All Components Bin-Matched
2. Bin Match with Noise
3. One Component Non-Bin-Matched

1.2.1 All Components Bin-Matched

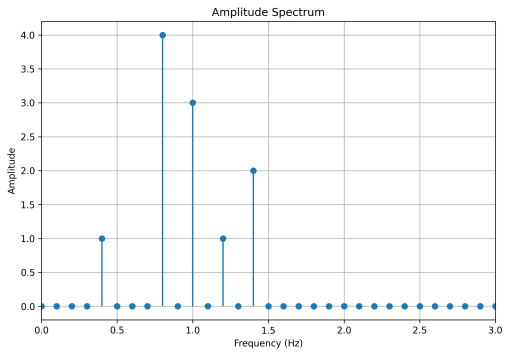
In this case, all input frequencies are chosen to fall on frequency bins, and the amplitude accuracy of the FFT output is verified for each component.

Table 1: FFT amplitude detection results for a multi-frequency bin-matched signal.

Input Frequency (Hz)	Input Amplitude	FFT Amplitude	Error (%)
0.40	1.0	1.000	0.00
0.80	4.0	4.000	0.00
1.00	3.0	3.000	0.00
1.20	1.0	1.000	0.00
1.40	2.0	2.000	0.00
		Overall Error	**0.00%**



(a) Multi-frequency input signal — all bin-matched.



(b) FFT amplitude spectrum — all components bin-matched.

Figure 4: Multi-frequency bin-matched case: input signal and corresponding FFT spectrum with zero error across all components.

As shown in Table 1 and Figure 4, all five frequency components are extracted with zero error. This confirms that when all signal frequencies fall on frequency bins, the FFT produces accurate results regardless of the number of frequency components present.

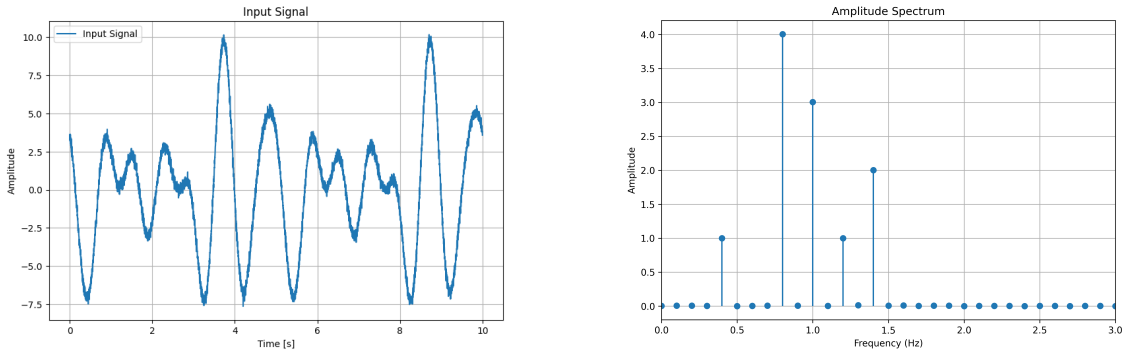
1.2.2 Bin Match with Noise

The same multi-frequency signal is used here, with the addition of white noise at an amplitude of 0.2. The aim is to assess whether the noise effect observed in the single-frequency case persists under multi-frequency conditions.

Table 2: FFT amplitude detection results for a multi-frequency bin-matched signal with added white noise.

Input Frequency (Hz)	Input Amplitude	FFT Amplitude	Error (%)
0.40	1.0	1.000	0.04
0.80	4.0	4.000	0.15
1.00	3.0	3.000	0.15
1.20	1.0	1.000	0.09
1.40	2.0	2.000	0.07
		Overall Error	**0.12%**

As shown in Table 2 and Figure 5, the introduction of white noise produces small errors across all frequency components. This result is consistent with the single-frequency case with noise, where the errors remain small relative to those caused by spectral leakage. White noise produces uniform amplitude perturbations across the entire frequency spectrum. Consequently, each amplitude component experiences the same level of disturbance regardless of how many signals are



(a) Multi-frequency input signal with white noise.

(b) FFT amplitude spectrum — bin match with noise.

Figure 5: Multi-frequency bin-matched case with added white noise: input signal and corresponding FFT spectrum.

superposed. Hence, the result is consistent in both cases.

1.2.3 One Component Non-Bin-Matched

In this case, the effect of a single frequency (within the multi-frequency signal) not falling on a frequency bin is investigated, specifically to check whether the resulting error is isolated to the non-matching frequency or propagates to neighbouring frequency components as well.

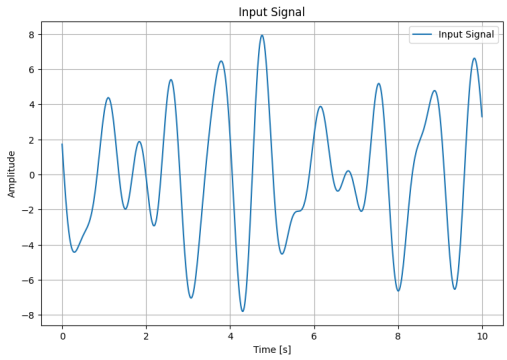
To investigate this, a multi-frequency signal identical to the previous case is used, with an additional component at 1.05 Hz introduced. Since the frequency bins are located at 1.0 Hz and 1.1 Hz, this frequency falls exactly midway between two adjacent bins, making it a worst-case spectral leakage scenario.

Table 3: FFT amplitude detection results for a multi-frequency signal with one non-bin-matched component at 1.05 Hz.

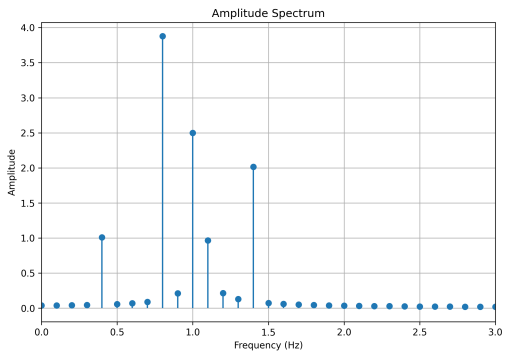
Input Frequency (Hz)	Input Amplitude	FFT Amplitude	Error (%)
0.40	1.0	1.0095	0.95
0.80	4.0	3.8793	3.02
1.00	3.0	2.4999	16.67
1.05	1.0	0.9654	3.45
1.10	1.0	0.2145	78.55
1.40	2.0	2.0152	0.76
		Overall Error	**17.23%**

As shown in Table 3 and Figure 6, spectral leakage from the 1.05 Hz component does not remain isolated to that frequency alone. It also corrupts the amplitudes of neighbouring frequency components, particularly 1.00 Hz and 1.10 Hz, which exhibit errors of 16.67% and 78.55% respectively.

A key observation is that the magnitude of this leakage-induced error decreases with increasing spectral distance from the non-bin-matched frequency. This spatial decay of leakage is also visible in the FFT spectrum, where the spurious disturbance diminishes progressively at frequencies further from 1.05 Hz. This behaviour denotes the importance of ensuring all frequency components in a multi-frequency signal fall on frequency bins.



(a) Multi-frequency input signal with one non-bin-matched component.



(b) FFT amplitude spectrum showing spectral leakage from the non-bin-matched component.

Figure 6: Multi-frequency case with one non-bin-matched component at 1.05 Hz: input signal and FFT spectrum illustrating leakage into neighbouring bins.

1.3 Discussions

The signal analysis conducted in this chapter provide important insights into the Fast Fourier Transform (FFT) and its limitations when applied to the extraction of signals. The study showed that when a signal frequency coincides exactly with an FFT frequency bin, the amplitude is identified without significant error, both for single-frequency and multi-frequency signals. However, when a frequency component does not fall on a bin (non-bin-matched case), spectral leakage introduces significant amplitude errors. In the single-frequency case, these errors can exceed 30%, depending on the relative position of the signal within the bin interval. In multi-frequency signals, spectral leakage from a single non-bin-matched component propagates to neighbouring frequencies, causing errors in their extracted amplitudes as well as to overall errors. White noise had a comparatively minor influence on amplitude extraction accuracy when frequencies are properly aligned with frequency bins, confirming that spectral leakage is the dominant source of error in FFT-based multi-frequency analysis.

The frequency resolution, determined by the length of the data record ($T = N/f_s$), plays an important role in reducing spectral leakage. If the length of the data record is increased, the possibility that signal frequencies fall close to a frequency bin in relative terms increases, thereby reducing leakage-induced errors. However, the sensitivity of the error to the exact relative position within a bin also increases with finer resolution. Hence, careful selection of simulation duration and excitation frequencies is needed.

These results illustrate the importance of frequency selection in the multi-frequency method for flutter derivative identification. When performing the multi-frequency simulations, prescribed motion frequencies should be chosen such that they coincide with FFT frequency bins for the chosen data length (total simulation time steps) and sampling frequency ($1/\Delta t$).